BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

Group ID: 25-26J-091

Project Title: Smart Real-Time Age, Gender, Emotion, and Behavior-Based Advertising System for Sri Lankan Malls

1. IT22109712: Interactive Game-Based Age, Gender, and Returning Customer Re-Identification using Temporal and Quality-Aware Face Embedding Similarity
2. IT22258526: A Context-Aware Real-Time Facial Emotion Recognition System for False Alert Reduction in Security Applications
3. IT22184030: Real-Time Context-Aware Behavior-Driven Engagement Analytics for Public Display Systems
4. IT22071934: Smart Billboard Advertisement Classification and Recommendation System Using Machine Learning and Context-Aware Targeting

		IT22109712	IT22258526	IT22184030	IT22071934
Data Pipeline	**Data Sources:**	The age and gender detection model uses publicly available datasets such as UTKFace and IMDB-WIKI. These datasets contain facial images labeled with age and gender, which are suitable for supervised learning.	The emotion detection component utilizes a publicly available Sri Lankan facial emotion dataset obtained from IEEE DataPort. This dataset contains facial images labeled with standard emotion categories such as happy, sad, angry, fear, and neutral.	This model uses simulated or derived datasets based on video streams. Features such as gaze direction, head orientation, and interaction are extracted using real-time data.	A custom dataset stored in MongoDB is used, containing advertisement metadata such as target age group, gender, mood, and category.
	Data Preprocessing:	Images are resized to a	Images are resized,	Video frames are	Data is cleaned,

		fixed resolution (e.g., 224x224) and normalized. Face detection is performed using MTCNN to extract relevant facial regions. Data augmentation techniques such as flipping, rotation, and brightness adjustment are applied to improve model generalization.	normalized, and augmented. Facial regions are extracted using MTCNN. Class imbalance is handled using weighted loss functions or oversampling techniques.	processed using OpenCV. YOLO is used for object detection, and tracking is done using SORT. Features such as attention percentage are calculated from detected behavior	missing values are handled, and categorical variables are standardized. Data is converted into structured formats such as DataFrames.
	Data storage and versioning	Training datasets are stored locally, while processed data can be managed using tools such as DVC for versioning. Git is used for code version control to ensure reproducibility.	Data is stored locally, and versioning is maintained using Git and optionally DVC for dataset tracking	Processed engagement data and logs are stored in MongoDB. Versioning is handled through Git	Advertisement data is stored in MongoDB. Versioning can be maintained using Git and database backups.
Model Development	**Model Selection**	A custom CNN-based age and gender classification model	A Custom CNN-based classification model is used to detect multiple	A combination of YOLO for object detection and a	A rule-based filtering system combined with

		built using TensorFlow/Keras is used for demographic prediction. The system additionally incorporates face embedding extraction and temporal quality-aware similarity modeling for returning customer identification. MTCNN is used for face detection and alignment, while cosine similarity, quality weighting, and temporal decay mechanisms are combined to improve identity consistency in unconstrained retail environments.	emotional states from facial expressions.The system additionally incorporates Context-aware emotion reasoning.	custom CNN for head orientation is used.	similarity-based methods (e.g., embeddings using Sentence Transformers) is used for recommendation.
	Model Training	The dataset is split into training, validation,	The dataset is split into training, validation, and	The model is evaluated based on	No traditional training is required

		and testing sets. The model is trained using categorical cross-entropy loss and evaluated using accuracy and F1-score. Hyperparameters such as learning rate and batch size are tuned. Face embeddings are generated using a pre-trained deep face recognition network and stored in MongoDB for returning customer analysis. Temporal-aware similarity and quality-aware weighting are applied during inference to reduce false identity assignments caused by low-quality detections and long-term	testing sets.The model is trained using labeled data with evaluation metrics such as accuracy, precision, recall, and F1-score. Cross-validation is used to improve robustness.	engagement accuracy and detection consistency rather than traditional classification metrics.	for rule-based systems. Evaluation is based on relevance and accuracy of recommendations.

		appearance variations.			
Model Integration	**Tools used**	Python, TensorFlow/Keras, OpenCV, MTCNN, ONNX Runtime, MongoDB, FastAPI	Python, TensorFlow/Keras, OpenCV	Python, OpenCV, SORT	Python, MongoDB, Pandas
Model Deployment	**Testing Environments**	Tested in a local environment using Python and Jupyter Notebook before integration into the system.	Tested locally with validation datasets before deployment.	Tested using recorded video streams and live camera feeds	Tested using simulated inputs (age, gender, emotion, engagement).
	Deployment Platform	Deployed on a local server or edge device for real-time inference.	Deployed on a local server integrated with the main system.	Deployed on an edge device (kiosk system) for real-time processing.	Deployed within the backend system connected to the database.
	Deployment Method	Exposed as part of a FastAPI-based real-time inference system capable of demographic prediction and returning customer identification using temporal and quality-	Integrated into the FastAPI backend for real-time emotion prediction.	Runs continuously as part of the real-time inference pipeline.	Provides recommendations through API endpoints in real time.

		aware face embedding similarity. The system operates in real time on a kiosk-based retail analytics environment.			
Future Enhancements	**Model Improvement:**	Improve accuracy using larger datasets and optimize model for faster inference using lightweight architectures. Enhance returning customer identification using adaptive temporal decay learning and multi-session embedding optimization while improving demographic fairness across different age and gender groups.	Incorporate more diverse datasets and improve real-time performance using optimized architectures.	Improve accuracy using advanced tracking methods and integrate additional behavioral features such as gesture recognition.	Implement machine learning-based recommendation systems and personalize ads using user history and behavior patterns.